

Your grade: 100%

Next item →

1. If the number of points inside a window is 16, the number of equations that you will get in the Lucas Kanade method will be?

1 / 1 point

- ☐ 4
- ☐ 8
- ☒ 16
- ☐ 32

☒ **Correct**
While adopting Lucas Kanade method, we get a large system of equations with the number of equations equal to the number of points inside the window. Therefore, we should 16 equations.

2. If we are able to estimate Optical Flow using Lucas Kanade method and matrix **A** has its relevant interpretation, which of the following conditions does **NOT** hold?

1 / 1 point

- ☐ $A^T A$ is invertible.
- ☐ λ_1 and λ_2 , the eigenvalues of $A^T A$, are $\lambda_1 \geq \lambda_2$ but not $\lambda_1 \gg \lambda_2$.
- ☒ $\det(A^T A) = 0$
- ☐ None of the above

☒ **Correct**
The first two conditions must hold for the optical flow estimation using Lucas Kanade method. The third condition directly violates the invertibility of $A^T A$. Therefore, c) is the correct option.

3. In which of the following cases can we use Lucas Kanade for optical flow estimation?

1 / 1 point

- ☐ Smooth Regions
- ☒ Textured region
- ☐ Edges
- ☐ All of the above

☒ **Correct**
A) The Lucas-Kanade method is not ideal for smooth or uniformly colored regions because it relies on local image intensity changes to compute optical flow. In smooth regions where there is little texture or gradient, it may have difficulty estimating motion accurately. Therefore, we can eliminate option A.

B) Textured regions with distinctive patterns and gradients are ideal for the Lucas-Kanade method. It works by tracking how these patterns move between frames, making it effective in regions with texture. This looks like the correct option.

C) The Lucas-Kanade method can not work with edges as in this case, one of the eigen values will be significantly larger than the other one as the gradient in one of the direction will be much greater than in the other direction. Therefore, we can eliminate option C.

Hence, option B is the correct response.